possible metrics; incidentally, this metric only works for centroid-based clustering algorithms, while it wouldn't be applicable to OPTICS or DBSCAN. A similar metric that would also work for those algorithms is the intra-cluster distance: the average pair-wise distance between points in the same cluster. Another interesting measure, useful when dealing with categorical features, is the total cohesion (similar to the distance to cluster center, but uses the cosine distance instead of the Euclidean distance).

In the rest of this section, we'll adopt the intra-cluster average distance as our metric. Now, however, it's time to reveal the *elbow method*, an empirical tool that's used, in clustering but also across machine learning, to spot the best value for hyper-parameters.

Figure 12.25 shows an example of applying the elbow method to our customers dataset, to decide what's the best number of clusters based on intra-cluster average distance; but, for instance, it would also be possible to use the same method to decide what's the best value for ε or `minPoints`, if we were using DBSCAN instead.